

BPAS: Blockchain-Assisted Privacy-Preserving Authentication System for Vehicular Ad-Hoc Networks

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Abstract—If all vehicles are connected together through a wireless communication channel, vehicular ad-hoc networks (VANETs) can support a wide range of real-time traffic information services such as intelligent routing, weather monitoring, emergency call. However, the accuracy and credibility of the transmitted messages among the VANETs is of paramount importance as life may depend on it. We introduce a novel framework called blockchain-assisted privacy-preserving authentication system (BPAS) that provides authentication automatically in VANETs and preserves vehicle privacy at the same time. This design is highly efficient and scalable. It does not require any online registration centre (except for system initialization and vehicle registration), and allows conditional tracing and dynamic revocation of misbehaving vehicles. We conduct an in-depth security analysis and a comprehensive performance evaluation (which is based on the Hyperledger Fabric platform) for our proposed framework. The results demonstrate that our framework is an efficient solution for the development of a decentralized authentication system in VANETs.

Index Terms—Authentication, blockchain, smart contract, vehicular ad-hoc networks.

1 INTRODUCTION

VEHICULAR ad-hoc networks (VANETs) are formed when the principle of mobile ad-hoc networks (MANETs) is applied to the domain of vehicles. In the VANET environment, vehicles are connected via wireless communication to make data exchange feasible. In this context, real-time information (such as traffic information, weather condition, road status) may help vehicles or traffic control centers take timely actions (e.g., collision avoidance, intelligent routing, traffic lighting) [1]. Therefore, VANETs have been receiving increasing attention from both academia and industry in the past few years. Fig. 1 depicts a set of complicated communication modes in a traditional VANET environment. The “smart” vehicles, equipped with on-board units (OBUs), are able to communicate with each other in vehicle-to-vehicle (V2V) mode, but also the road side infras-

tructures (such as road side units (RSU)) through vehicle-to-infrastructure (V2I) mode. Specifically, both communication modes follow the dedicated short range communication (DSRC) protocol that supports data exchange over a short range (generally within a few meters) even when vehicles are moving in high speed.

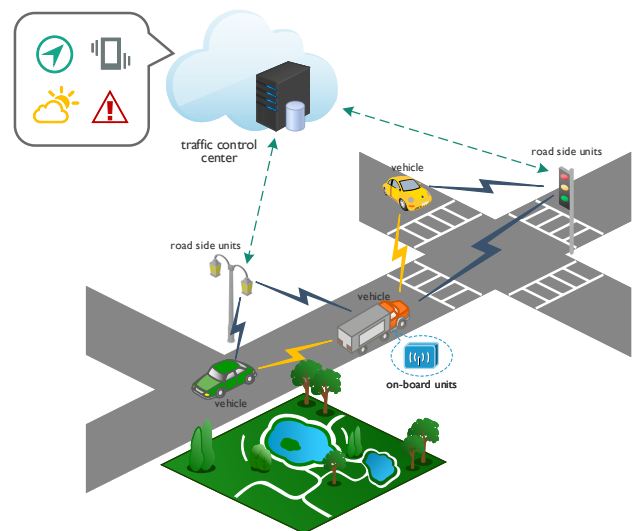


Fig. 1: A typical structure of a vehicular ad-hoc network

In this way, a vehicle can act as an information collector. It informs others of what it perceives to help vehicles dynamically update driving route for fuel saving, congestion or accident avoidance. Besides, an RSU also relays these messages to the traffic control centre for further traffic management and consultation services. For example, in the eSafety Support project [2], an emergency call is made once in-vehicle sensors detect that an accident has occurred. Such

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information exchanged in the VANET context must be accurate, trustful and truthful, as lives and important decisions may depend on it. However, due to the openness of wireless communications, malicious entities may intercept, relay, and even tamper the transmitted messages. If an attacker reports fraudulent information about the traffic condition or vehicle position, this may result in bad consequences such as traffic chaotic or road accidents [3]–[5]. Furthermore, similar to other networks, the nodes of VANETs may misbehave toward those honest ones.

Therefore, secure authentication on the transmitted messages is an important requirement in VANETs. Another essential issue for VANETs is conditional anonymity, i.e., a vehicle's private information (e.g., physical serial number) should only be visible to trusted authority and any third-party observer cannot violate driver privacy [6], [7]. This feature emphasizes both privacy and accountability when fake messages result in crimes or accidents. Over the past few years, many privacy-preserving authentication protocols [8]–[13] have been proposed in the literature. However, we observe that the previously proposed protocols are highly dependent on a centralized server. For example, public key infrastructure (PKI)-based solutions require a certificate authority whereas ID-based solutions rely on a key generation centre. The former suffers from cumbersome certificate management while the latter is vulnerable to the key escrow problem. A hybrid-based scheme combining the two types of the solutions may mitigate the drawbacks, but it is still not scalable enough in practice.

Blockchain technology, originally proposed in [14], has been seen as a potential solution to bring “trust” and “auto-check” to VANET. A blockchain platform is an append-only database maintained by the nodes of a peer-to-peer (P2P) network, where the nodes are geographically dispersed but being equally privileged participants in the application. Each node engages in the routing process of the entire network, maintaining the connections to neighboring peers, the propagation and verification of signed messages, as well as the synchronization of data blocks (that are chained by hash pointers in chronological order and synchronized using a cost-effective consensus mechanism). The “flat” topology of blockchain, therefore, offers network users autonomous, decentralized, immutable and contractual “benefits” [15]. With the purpose of injecting auto-trust into VANET, blockchain may be considered as a powerful and scalable tool that can automatically check message credibility, monitor vehicle behavior and further trace immutable communication record. This paper is motivated by the security and privacy challenges of VANETs and explores the question of “*could we build a blockchain-assisted distributed authentication system for VANETs with privacy-preserving feature?*”.

1.1 Our Research Contributions

We summarize our key research contributions as follows.

- By leveraging the blockchain technology, we design a novel authentication system BPAS for VANETs to guarantee that the transmitted messages are verified credibly in the absence of a centralized third party.
- We perform a comprehensive security analysis to show that the proposed BPAS framework satisfies security

and privacy requirements of VANETs, specifically, the conditional privacy preservation.

- Finally, we implement a prototype of BPAS using smart contracts and we deploy it on the well-known consortium blockchain platform, Hyperledger Fabric. The performance evaluation results obtained demonstrate that BPAS has great potential to become a practical component for VANETs.

1.2 Organization of the Paper

The rest of the paper is organized as follows. Section 2 discusses some related works on privacy-preserving authentication schemes for VANETs. In Section 3, we define the system model and design goals. We present the fundamental building blocks of BPAS in Section 4. The proposed blockchain-assisted privacy-preserving authentication system for VANETs is described in Section 5. Section 6 focuses on the security of BPAS. In Section 7, we provide an experimental evaluation of the proposed system. Finally, we make some concluding remarks in Section 8.

2 RELATED WORKS

In recent years, many researchers have focused on security and privacy issues in VANETs [16]–[24]. Raya and Hubaux [17] employed PKI to meet the authentication requirement for VANETs. They used anonymous certificate (AC), issued by a trust third authority, to hide a vehicle's real identity. Therefore, each time the vehicle communicates with others, the validity of the transmitted messages and the privacy of the sender can be guaranteed simultaneously. After that, Lu et al. [19] enhanced the unlinkability property by using temporary AC for each session. Calandriello et al. [18] simplified the certificate management procedure via a group signature and baseline pseudonym mechanism. Wasef et al. [20] presented a scheme called expedite message authentication protocol (EMAP) which adopts PKI for vehicle authentication and hash-based authentication code to optimize the integrity check process. However, all the PKI-based authentication schemes suffer from similar weaknesses: 1) The vehicle needs to hold a fair number of secret keys and certificates in order to protect its privacy; 2) The trusted third-party requires a large space to store all the certificates; 3) The management of these certificates (e.g., query or revocation) incurs high computation and communication costs.

Another option for the authentication process is identity-based cryptography (IBC), which can efficiently reduce the communication cost of VANETs. IBC was designed by Shamir [25] where each user private key is issued by the key generation center (KGC) based on his/her identity (e.g., e-mail address) which is also his/her public key. The natural connection between identity and public key makes entity identification easier. Therefore, Zhang et al. [21] employed IBC in VANETs for lightweight message authentication. Chim et al. [22] enhanced the privacy of Zhang et al.'s scheme by using two shared secrets, but their method is still vulnerable to impersonation attack as pointed out by Horng et al. [9]. To overcome the security flaw of the previous works, He et al. [24] and Li et al. [10] applied IBC

for secure and extendable communications in VANETs. Although these schemes make better improvement compared with PKI-based frameworks, they suffer from the inherent drawbacks of the key escrow issue and the need of secure communication channel.

Recently, with the emergence of blockchain technology, some research efforts have been exploring to apply it in VANETs to build a decentralized trust model. For example, in [26], Rowan et al. leveraged blockchain-based PKI with physical side-channel for secure V2V communication. However, their scheme suffers from some security issues due to the requirement of autonomous vehicle. Dorri et al. [27] proposed another privacy-preserving authentication based on blockchain and changeable public keys which also suffers from limitations, such as membership management and scalability. Lu et al. [28] and Kchaou et al. [29] used blockchain to optimize the VANET trust management framework and designed the privacy-aware reputation models. The transaction in both schemes are assumed to securely record the events around the vehicles. Such events could act as persistent evidence later for vehicle's reputation evaluation. Although their schemes support strong accountability, they cannot prevent malicious behaviors beforehand.

3 PROBLEM FORMULATION

3.1 System Architecture

Fig. 2 describes the network model of our proposed BPAS framework which consists of three components, namely, the upper layer (i.e., trusted authority), the bottom layer (i.e., vehicle and road side unit), and the extended layer (i.e., smart contract and blockchain). We are going to define five participants below.

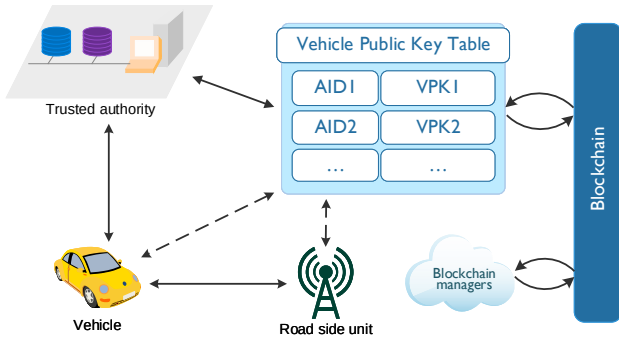


Fig. 2: The system model of our deigned framework¹

- **Trusted authority (TA):** It is the trusted third party being responsible for system initialization, smart contract deployment, vehicle registration and revocation. It is generally assumed that the TA has considerable communication and computation capabilities, and will not collude with other parties.
- **Vehicle:** It performs services for its driver and is equipped with a tamper-proof device OBU. The information preloaded in the OBU is assumed to be safe

1. The solid line indicates the direct communication and the dotted line indicates the indirect communication with the assistance of transactions and blockchain managers

from malicious attacks. Furthermore, the vehicle communicates with other parties wirelessly via the OBU.

- **Road side unit (RSU):** It is the infrastructure located at the roadside and can communicate wirelessly with vehicles within a specific range. To be more specific, it can receive instant messages from vehicles, verify, and relay them to the traffic management center or to other vehicles.
- **Smart contract:** It loads a vehicle public key table (VPKT) to aggregate references to all of the vehicles' "identity-public key" relationships, thus providing automatic and timely feedback on vehicles' public key queries. In particularly, they are on-chain codes that are stored in the blockchain, executed and verified by the blockchain managers. Therefore, the smart contract ensures reliable computed results.
- **Blockchain:** It is instanced using a consortium blockchain as the decentralized underlying architecture of BPAS. It securely handles the states of transactions and smart contracts among several consensus nodes (named **blockchain managers**) based on the consensus algorithm. Permitted nodes could query these states at any time to obtain a result verified by the majority of the blockchain managers.

We assume that the TA authenticates the vehicles' owners and preloads authenticators in the OBU offline. The vehicles and RSUs communicate with each other based on the DSRC protocol. Only the TA can deploy, update and revoke the smart contracts via transactions, and any permitted parties can access information using transactions. Synchronously, blockchain managers can verify all the transactions and the new states of smart contracts, and can upload them in the blockchain based on the consensus mechanism.

3.2 Design Goals

Following the latest research efforts [15], [24], [27], a blockchain-assisted authentication system for VANETs should satisfy the following security requirements.

- **Single registration:** For convenience, an authentication system for VANETs should support the single registration, where each vehicle needs to register only once before it sends the message to other parties on the road.
- **Message authentication:** To ensure the credibility of received messages, the RSUs or vehicles should be able to authenticate the messages, involving the validity of originators' identities, the integrity and timeliness of messages.
- **Privacy preservation:** The real identity of a vehicle should be invisible to the RSU or other vehicles, and any adversary should not be able to get the real identity by analyzing the intercepted transcripts.
- **Traceability:** There should be an effective way for the TA to trace a vehicle's real identity when malicious behaviors have occurred, for example, fake messages are sent to mislead others.
- **Three-factor security:** To protect the user's privacy, an authentication system for VANETs should meet the three-factor security, i.e., an adversary can not extract any information about the three secret authenticators.

- *No online registration center*: To relieve the overhead of the registration center, an authentication system should not rely on the online registration center during the authentication phase.
- *Resistance to cyber attacks*: Generally, a blockchain-assisted authentication system for VANETs should be able to resist the offline password guessing attack, replay attack, vehicle impersonation attack, and distributed denial of service attack.

4 BUILDING BLOCKS

Here, we introduce some fundamental primitives used in BPAS and we present a vehicle public key table deployed with fine-grained access control.

4.1 Fuzzy Extractor

In order to enhance the security of authentication, we leverage fuzzy extractor presented by Dodis et al. [30] as a fundamental component. It is one of the most popular biometrics extraction techniques based on a fuzzy extractor. Specifically, the fuzzy extractor is formalized by a tuple of $\{m_e, l, \tau, \varepsilon\}$ and two algorithms $\{\text{Gen}, \text{Rep}\}$, where m_e denotes the min-entropy of collected biometrics data distribution, l denotes the fixed length of output, τ and ε define the tolerable threshold of sample distance and statistical distance respectively. If we assume that BiO is a biometrics sample, then the algorithms $\{\text{Gen}, \text{Rep}\}$ are defined as follows:

- $\text{Gen}(BiO) = (\sigma, \rho)$: It denotes a probability extraction function with the input of a biometrics sample, and the outputs of a high-entropy secret string $\sigma \in \{0, 1\}^l$ and a public string ρ . The statistical distance, denoted as $SD((\sigma, \rho), (\theta, \rho))$, is assumed to be smaller than the pre-defined threshold ε , where θ is an l -bit length of random string.
- $\text{Rep}(BiO', \rho) = \sigma$: It denotes a deterministic retrieve function with the inputs of a biometrics sample, the public string. It is assumed to be error-tolerated that the secret string σ could be recovered correctly if and only if the sample distance $\text{dis}(BiO, BiO') < \tau$.

It should be noted that:

- 1) If the BiO and BiO' are collected from the same user, there is a high possibility that the distance between them is lower than the threshold, i.e., $\Pr[\text{dis}(BiO, BiO') < \tau] > 1 - \varepsilon_{fn}$, where ε_{fn} is the possibility of a “false negative”.
- 2) If the BiO^1 and BiO^2 are collected from different users, there is a high possibility that the distance between them is far larger than the threshold, i.e., $\Pr[\text{dis}(BiO^1, BiO^2) > \tau] > 1 - \varepsilon_{fp}$, where ε_{fp} is the possibility of a “false positive”.

4.2 Attribute-Based Encryption (ABE)

ABE is a scheme where an authority issues an attribute secret key for a user based on a set of attributes that can be used to identify this user. In addition to the general features of the public key encryption (e.g., correctness, message

integrity and confidentiality), it can further provide fine-grained access control. We use the well-known construction [31] for the BPAS. Let Ψ denote the access tree where each non-leaf node represents a threshold gate and a leaf node is described as an attribute. Let $\mathcal{L}$ be the set of leaf nodes in Ψ with the length of ℓ . Let $\text{att}(x)$ denote the function that returns the attribute associated with leaf node $x \in \mathcal{L}$ and t_x denote its threshold value. The attribute-based encryption scheme is defined as follows.

- **ABE.TSetup**: This algorithm outputs the master secret key MSK and the public parameters $ABEParams = \{\mathbb{G}_0, \mathbb{G}_1, p, \mathcal{H}, g, MPK\}$, where $\mathbb{G}_0, \mathbb{G}_1$ are two cyclic groups with same order of prime number p and satisfy a bilinear pairing mapping $e : \mathbb{G}_0 \times \mathbb{G}_0 \rightarrow \mathbb{G}_1$, $\mathcal{H}$ is a cryptographic hash function defined as $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{G}_0$, g is a random generator of $\mathbb{G}_0$, $MSK = \{\beta, g^\alpha\}$ is the master secret key defined by two random numbers $\alpha, \beta \in \mathbb{Z}_p$, $MPK = \{h, f, \Gamma\}$ is the master public key defined by the equations of $h = g^\beta, f = g^{\beta^{-1}}, \Gamma = e(g, g)^\alpha$. Remark that the master secret key MSK should be kept securely.
- **ABE.AttrEnc**: Upon receiving the inputs of message M along with the master public key MPK and the access tree structure Ψ , this algorithm outputs the ciphertexts $CT = (\Psi, R, T, S_1, \dots, S_\ell, P_1, \dots, P_\ell)$, where $R = M \cdot \Gamma^r, T = h^r, S_i = g^{p_i(0)}, P_i = \mathcal{H}(\text{att}(i))^{p_i(0)}$ for all the $i \in \mathcal{L}$, with the unique polynomial p_x for each node x (including the leaves) in Ψ and random number $r \in \mathbb{Z}_p$ (It is worth noting that for the root $\mathcal{R}$ of Ψ , we set $p_{\mathcal{R}}(0) = r$).
- **ABE.AKeyGen**: Upon receiving the inputs of attribute set $\mathbb{A}$ and master secret key MSK , this algorithm outputs the secret key $SK_{\mathbb{A}} = \{K_0, \{K_i, K'_i | i \in \mathbb{A}\}\}$ where $K_0 = g^{(\alpha+\mu) \cdot \beta^{-1}}, K_i = g^\mu \cdot \mathcal{H}(i)^{\mu_i}, K'_i = g^{\mu_i}$ for all $i \in \mathbb{A}$, with the random numbers $\mu, \mu_1, \dots, \mu_{|\mathbb{A}|} \in \mathbb{Z}_p$.
- **ABE.AKeyDel**: Upon receiving the inputs of secret key $SK_{\mathbb{A}}$ along with a subset of attributes $\mathcal{A} \subseteq \mathbb{A}$, this algorithm delegates a new secret key $SK_{\mathcal{A}} = \{\tilde{K}_0, \{\tilde{K}_i, \tilde{K}'_i | i \in \mathcal{A}\}\}$ where $\tilde{K}_0 = K_0 \cdot f^{\tilde{\mu}}, \tilde{K}_i = K_i \cdot g^{\tilde{\mu}} \cdot \mathcal{H}(i)^{\tilde{\mu}_i}, \tilde{K}'_i = K'_i \cdot g^{\tilde{\mu}_i}$ for all the $i \in \mathcal{A}$, with the random numbers $\tilde{\mu}, \tilde{\mu}_1, \dots, \tilde{\mu}_{|\mathcal{A}|} \in \mathbb{Z}_p$.
- **ABE.Decrypt**: Upon receiving the ciphertext CT along with the secret key $SK_{\mathcal{A}}$ associated with the attribute set $\mathcal{A}$ and a node $x \in \Psi$, the recursive algorithm $\text{DecryptNode}(CT, SK_{\mathcal{A}}, x)$ is defined as follows:
 - 1) If $x \in \mathcal{L}$, i.e., x is a leaf node, let $i = \text{att}(x)$, then $\text{DecryptNode}(CT, SK_{\mathcal{A}}, x) =$

$$\begin{cases} \frac{e(K_i, S_x)}{e(K'_i, P_x)} e(g, g)^{\mu \cdot p_x(0)}, & \text{if } i \in \mathcal{A}; \\ \perp, & \text{if } i \notin \mathcal{A}. \end{cases}$$
 - 2) If $x \notin \mathcal{L}$, i.e., x is a non-leaf node, let ζ denote the children of x and $\mathcal{O}_\zeta$ be the output of $\text{DecryptNode}(CT, SK_{\mathcal{A}}, \zeta)$, $\mathcal{A}_x$ be an arbitrary t_x -sized sub-set of child ζ such that $\mathcal{O}_\zeta \neq \perp$, then $\text{DecryptNode}(CT, SK_{\mathcal{A}}, x) =$

$$\begin{cases} \perp, & \text{if no such } \mathcal{A}_x \text{ exists;} \\ \prod_{\zeta \in \mathcal{A}_x} \mathcal{O}_\zeta^{\lambda_{i, \mathcal{A}_x}(0)} = e(g, g)^{\mu \cdot p_x(0)}, & \text{otherwise.} \end{cases}$$

where $i = \text{index}(\zeta)$, $\mathcal{A}'_x = \{\text{index}(\zeta) : \zeta \in \mathcal{A}_x\}$ and the Lagrange coefficient $\lambda_{i,\mathcal{A}'_x}(0) = \prod_{j \in \mathcal{A}'_x, j \neq i} \frac{-j}{i-j}$. The result can be deduced using polynomial interpolation.

Now the ciphertext can be decrypted by calling $\delta = \text{DecryptNode}(\text{CT}, SK_{\mathcal{A}}, \mathcal{R})$ on the root of Ψ . Once the tree is satisfied by $\mathcal{A}$, i.e., $\delta = e(g, g)^{\mu \cdot \mathcal{P}_{\mathcal{A}}(0)} = e(g, g)^{\mu \cdot r}$, then we have

$$M = \frac{R \cdot \delta}{e(T, K_0)} = \frac{M \cdot \Gamma^r \cdot \delta}{e(h^r, g^{(\alpha+\mu) \cdot \beta^{-1}})}$$

The security of this scheme has been proven in the generic group model. A more in-depth description is given in [31].

4.3 Blockchain and Smart Contract

Blockchain is the core data storage structure of BPAS. In its most general form, blockchain can be seen as a distributed, transparent, and secure data ledger, where the data (as well as any change of the data) can be recorded in an append-only chain of blocks chronologically. Based on the different requirements of access control, blockchain platforms can provide three types of application instances: private blockchain, consortium blockchain and public blockchain. In BPAS, we choose the open source platform Hyperledger Fabric, which is one of the classic consortium blockchain applications, as the basic network architecture. The reason for the choice is because the Hyperledger Fabric can handle transactions efficiently and support Turing-complete contracts. Furthermore, the Hyperledger Fabric naturally enables access control strategies (authorized endorser peers can keep the chain codes while both committer and endorser peers can synchronize the ledger)

Smart contract, firstly coined by Nick Szabo in the 1990s, “is a computerized transaction protocol that executes the terms of a contract. The general objective is to satisfy common contractual conditions”. Generally, it, coupled with the blockchain technology, provides some attracting features, i.e., self-executing, immutable, self-verifying, auto-enforcing, and decentralization. That is why smart contract can be auto-executed and deployed in decentralized network. In our BPAS design, we make use of smart contract which provides application binary interfaces (ABIs) for vehicle public key table (VPKT) management services. These ABIs support inserting, uploading, revoking of the public keys. **Algorithm 1~4** shows the smart contracts that are deployed in BPAS. The **Algorithm 1** declares the structure of elements in VPKT and the transactions that are initialized in the Hyperledger Fabric to specific the query format; the **Algorithm 2** declares the functionality when the TA intends to update some vehicles’ public keys that already exist in the VPKT; the **Algorithm 3** shows the insert function for the TA when new vehicle has been registered and a new item is waiting to be added in VPKT; finally, the **Algorithm 4** defines the ABI for the TA to remove an item of VPKT when the associating vehicle waits to be revoked.

We note that the VPKT maintained in the smart contract can be queried with suitable pre-defined permissions. As for BPAS, we require that only managers and TA can query the VPKT, while other entities (such as RSUs or vehicles)

Algorithm 1. Vehicle public key table (VPKT) initialization

% This declares the structure of elements in VPKT.

asset VPKT identified by AID {

o string AID;
o address VPK;

}

% This defines the structure of the transactions in the Hyperledger

transaction insertTransaction {

o string newAID;
o address newVPK;

}

transaction updateTransaction {

-- > VPKT asset; % Loading the VPKT.
o address newVPK;

}

transaction removeTransaction {

o string AID;

}

Algorithm 2. Update VPKT

function updateVPKT (tx) {

% This ABI is invoked by the TA to update VPKT.

const assetRegistry = await getAssetRegistry(VPKT);

if Exist(VPKT, tx.AID) then {

tx.asset.VPK = tx.newVPK;

await assetRegistry.update(entry);

return *succ*; }

else

return *err*;

}

can submit general transactions to managers when asking for a public key and the latter (acting as the distributed agents) search the result for them. This setting fully relies on the characteristic of the consortium blockchain, which can provide fast query of the vehicle’s public key and protect the privacy.

5 DESIGN OF BPAS

This section mainly introduces the design of our BPAS which consists of five modules: system initialization, smart contract deployment, vehicle registration, login and message authentication, and vehicle revocation.

5.1 System Initialization

In this phase, the TA generates the system parameters and initializes a consortium blockchain.

- **ECC initialization:** TA generates the system parameters of $\text{ECPParams} = \{q, a, b, n, P, P_{pub}, \mathcal{H}_1, \mathcal{H}_2\}$ where q , and $a, b \in \mathbb{F}_q$ (satisfying $4a^3 + 27b^2 \neq 0$) are the parameters with which to define a non-singular elliptic curve $\mathbb{E}(q)$ by the equation $y^2 = x^3 + ax + b \pmod{q}$; P is the generator of an additive group $\mathbb{G}_{EC}$ with the order of n (it is worth noting that $\mathbb{G}_{EC}$ consists of all the points over $\mathbb{E}$ and the point at infinity $\mathcal{O}$.); $P_{pub} = s \cdot P$ is the system public key associating with the system secret key $s \in \mathbb{Z}_n^*$; $\mathcal{H}_1 : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$, $\mathcal{H}_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_n$ are two cryptographic hash functions with a fixed range of output.
- **ABE Initialization:** TA defines a suitable access structure Ψ and attribute set $\mathbb{A}$, then it invokes ABE.TSetup and ABE.AKeyGen to generate

Algorithm 3. Insert VPKT

```
function insertVPKT(tx) {
  % This ABI is invoked by the TA to insert a new VPK.
  const assetRegistry = await getAssetRegistry(VPKT);
  if Exist(VPKT, tx.AID) then
    return err;
  var entry = newResource(VPKT, tx.newAID);
  entry.APK = tx.newVPK; await assetRegistry.add(entry);
  return succ;
}
```

Algorithm 4. Revoke VPKT

```
function revokeVPKT(tx) {
  % This ABI is invoked by the TA to revoke a vehicle.
  const assetRegistry = await getAssetRegistry(VPKT);
  if Exist(VPKT[i].AID == tx.AID) then {
    assetRegistry.remove(VPKT[i].AID);
    return succ; }
  else
    return err;
}
```

ABEParams = $\{\mathbb{G}_0, \mathbb{G}_1, p, \mathcal{H}, g, MPK\}$, the master secret key MSK and the master attribute key SK_A based on Ψ and $\mathbb{A}$. When each blockchain manager registers in this system, TA will invoke ABE.AKeyDel to issue a sub-key $SK_{\mathcal{A}}$ based on its entity's attributes $\mathcal{A}$.

- **Blockchain initialization:** TA starts up a consortium blockchain among the preset network nodes following the PBFT (i.e., Practical Byzantine Fault Tolerance) consensus mechanism to maintain the blockchain. It should be noted that TA has authenticated all the blockchain managers in advance and authorized them to engage in the consensus process. In addition, it will deploy an access control list in the Hyperledger Fabric, such as READ, RECORD.

5.2 Smart Contract Deployment

The TA takes the inputs of smart contract drafts (designed in Section 4.3), compiles and deploys them into the blockchain. After being verified by the blockchain managers, these smart contracts get their unique addresses and can be invoked using transactions with suitable permissions.

5.3 Vehicle Registration

Each vehicle and its owner need to register with the TA to get their secret authentication factors in this phase. We note that this phase is executed within a secure channel.

- 1) First, the owner chooses a physical identity VID , a password pw and imprints his/her biometrics BiO at the sensor to extract $(\sigma, \rho) \leftarrow \text{Gen}(BiO)$. Finally he/she submits VID together with $\{pw, \sigma, \rho\}$ to the TA.
- 2) Second, TA computes the blinding identity as $AID = VID \oplus \mathcal{H}_1(sk \cdot P_{pub})$ where $sk = \mathcal{H}_2(VID||s)$ is the secret key for this vehicle. TA further computes $\mathcal{K} = \mathcal{H}_1(VID||pw||\sigma)$, $VPK = sk \cdot P$.
- 3) Finally, TA loads $\{\mathcal{K}, \rho, \text{ABEParams}, \text{ECPParams}, \Psi, VID, AID\}$ in the vehicle's OBU (which is assumed to be temper-proof). Furthermore, TA uploads $\{AID, VPK\}$

to the smart contract via the insert ABI, i.e., $\text{insertVPKT}(AID, VPK)$, by using a private transaction.

5.4 Login and Authentication

When a vehicle intends to send some traffic messages to nearby RSUs and vehicles, the following steps are executed.

- 1) The vehicle's owner inputs pw' and imprints his/her biometrics BiO' at the sensor. The OBU computes $\sigma' \leftarrow \text{Rep}(BiO', \rho)$ and checks if $\mathcal{K} = \mathcal{H}_1(VID||pw'||\sigma')$ holds. If this verification fails, the OBU rejects the request. Otherwise, it encrypts the blinding identity AID as $\Upsilon \leftarrow \text{ABE.AttrEnc}(AID)$ to guarantee that only the blockchain managers could decrypt it. It further generates a random number $r \in \mathbb{Z}_n$ and computes $R = r \cdot P$, $\alpha = \mathcal{H}_2(\Upsilon||R||M||T_1)$ and $\omega = r + \alpha \cdot sk \mod n$, where M is the instant message being valid during the timestamp T_1 . Finally, the OBU broadcasts the message $\{\Upsilon, M, R, T_1, \omega\}$ to nearby RSUs and vehicles.
- 2) The receiver (an RSU or a vehicle) checks the freshness of the received message using the timestamp T_1 . If it is valid, the receiver launches a transaction with Υ to the blockchain managers for the request of associating public key.
- 3) The blockchain managers (who satisfy the attributes that appear in Ψ) can invoke the ABE.Decrypt to decrypt it to AID . They further search the VPKT identified by AID to get the suitable public key VPK or *err*.
- 4) Upon seeing the VPK from the blockchain managers, the receiver computes $\alpha' = \mathcal{H}_2(\Upsilon||R||M||T_1)$ and accepts the message if and only if $\omega \cdot P$ equals to $R + \alpha' \cdot VPK$; otherwise, it rejects this message.

Due to $VPK = sk \cdot P$, $\omega = r + \alpha \cdot sk \mod q$, we have

$$\omega \cdot P = (r + \alpha \cdot sk) \cdot P = R + \alpha \cdot VPK$$

Therefore, the correctness of the authentication process is proved. Once the message being modified, the RSUs or vehicles can notice and abort it immediately.

5.5 Vehicle Revocation

When a vehicle leaves this area or waits to be scrapped, the TA needs to terminate the current state recorded in the smart contract. As described above, BPAS relies on VPKT to manage the public keys, thus, it is easy for TA to revoke a vehicle via the ABI of $\text{revokeVPKT}(AID)$ to delete the associated tuple. Since that information in the OBU cannot be modified after initialization, if the vehicle's account identity or password needs to be changed, TA revokes the old one and starts a new registration session.

To avoid malicious deletion or amendment, we follow the idea of mandatory access control (MAC) so as to ensure that only the TA can change the state of VPKT. These administration principles can be extended by adding other complex and flexible policies to enforce authorized behaviors.

6 SECURITY ANALYSIS

We analyze the security of our proposed BPAS in this section. In particular, we show that BPAS satisfies all the security requirements listed in Section 3.2.

- **Single registration:** According to the description of BPAS, the TA generates authentication information for a vehicle during the registration phase, after which each vehicle can be authenticated by RSUs or other vehicles without TA. Therefore, BPAS can meet the requirement of single registration.
- **Message authentication:** The RSU or the vehicle can check the validity of received messages following the equation of $\omega \cdot P = R + \alpha \cdot VPK$, where $\alpha = \mathcal{H}_2(\mathcal{Y}||R||M||T_1)$. Based on the collision resistance of hash functions and the elliptic curve discrete logarithm problem, no probabilistic polynomial time adversary can forge a valid authentication transcript, and any modification of the message will be detected by the RSUs or other vehicles. Therefore, BPAS supports reliable authentication on the transmitted messages.
- **Privacy preservation:** During the authentication phase, BPAS makes use of ABE to protect the vehicle's real identity VID . Based on the security assumptions of ABE, we note that none of the message receivers can observe the real identity or link two messages to one entity. Further, the blockchain managers act as the mixers who decrypt the ciphertext to get the masked identity AID for the public key query. Therefore, the unlinkability is satisfied. Finally, since $AID = VID \oplus \mathcal{H}_1(sk \cdot s \cdot P)$ where the blinding factor is a Diffie-Hellman tuple, due to the CDH problem, we conclude that no malicious entities (including RSUs, other vehicles or third parties) can get workable information about the real identity. Therefore, BPAS can preserve a vehicle's privacy efficiently.
- **Traceability:** In the design of the smart contracts, the identity information stored in the smart contracts is masked value $AID = VID \oplus \mathcal{H}_1(sk \cdot s \cdot P)$, where sk is the secret key of vehicle and s is the master key for TA. This means that only the TA and the associated vehicle can see the real identity efficiently. Therefore, in the event of a bad behavior, the TA can easily discover the real identity based on the records stored in the blockchain. Furthermore, this accountability is credible due to the data integrity provided by the blockchain. Therefore, BPAS could provide traceability.
- **Three-factor security:** The OBU equipped in the vehicle is assumed to be a tamper-proof device and its information is never lost. Thus, we just consider the other two factors, i.e., password and biometrics. We note that what the vehicle broadcasts does not include any information about them ($\mathcal{Y}$ is the encrypted value about the identity, M is the broadcast message, R is randomly generated element, T_1 is the timestamp and ω is equal to $r + sk \cdot \mathcal{H}_2(\mathcal{Y}||R||M||T_1) \bmod n$). Therefore, even by eavesdropping the traffic, an adversary cannot guess the password pw^* or extract the biometrics BiO , that is, BPAS provides three-factor security.
- **No online registration center:** According to the description of BPAS, the RSUs or vehicles check the valid-

ity of the received messages by verifying the signature on them with the assistance of smart contracts, which are deployed and executed within the decentralized blockchain. We can see that the TA is not involved throughout the authentication phase. Therefore, BPAS can achieve authentication in the absence of the online registration center.

- **Resistance to offline password guessing attack:** Suppose that there is an adversary who can eavesdrop on the communication network and observe the message $\{\mathcal{Y}, M, R, T_1, \omega\}$. However as previously analyzed, these intercepted messages do not contain any information about the password string. In other words, there is no way for the adversary to check the correctness of his/her guess about the password pw . Therefore, BPAS can resist the offline password guessing attack.
- **Resistance to replay attack:** During the login and authentication phase, a vehicle broadcasts $\{\mathcal{Y}, M, R, T_1, \omega\}$ for the messages' transmission and authentication. In order to guarantee the freshness of the messages, a timestamp must be involved. As previously analyzed, $\mathcal{Y}$ is a ciphertext generated from $ABE.AttrEnc(AID)$, $R = r \cdot P$, $\omega = r + sk \cdot \alpha$ and $\alpha = \mathcal{H}_2(\mathcal{Y}||R||M||T_1)$. Due to the indistinguishability property of ABE, the randomness of r , the freshness of T_1 and the security assumptions of hash function, the RSUs and other vehicles can easily detect any replay behavior by checking the validity of received message. Therefore, BPAS can withstand the replay attack.
- **Resistance to vehicle impersonation attack:** In the login phase, the OBU identifies the owner by verifying $\mathcal{K} = \mathcal{H}_1(VID||pw||\sigma)$ where σ is the secret string extracted from owner's biometrics by $Gen(\cdot)$ function. As we have analyzed before, BPAS satisfies three-factor security because there is no probabilistic polynomial time adversary that can create a valid login request. On the other hand, a valid broadcast message needs to be signed by the secret key sk pre-loaded in OBU, the adversary cannot impersonate a vehicle to send a message to other parties. Therefore, BPAS can hold against the vehicle impersonation attack.
- **Resistance to distributed denial of service attack:** BPAS inherits blockchain's resistance to distributed denial of service (DDoS) attack, which means that any modification about the smart contracts and transactions can be prevented and any illegal transactions will not be recorded into the blockchain. Furthermore, the query on the vehicle's public key is answered by the ABI in smart contract which is executed within the blockchain automatically even when some points are blocked in the network. Therefore, BPAS can resist the DDoS attack.

7 PERFORMANCE ANALYSIS

First, we evaluate the time overhead of cryptographic operations. Here, we conduct our performance evaluation on a laptop running Ubuntu 18.04 OS and equipped with an Intel (R) Core (TM) i7-6700 CPU @3.40GHz and 2GB RAM. The primitive cryptographic operations such as point addition, point multiplication, hash functions are implemented using

the relic library [32]. The ABE scheme is instantiated based on the cpabe-toolkit [33]

Table 1 and Table 2 present the results obtained. Here, we can approximately calculate the execution times related to the cryptographic operations during a successful authentication. It should be noted that the TA is offline for the authentication phase.

- 1) When the owner logs into the vehicle, one deterministic retrieve function, one hash functions are required for identification.
- 2) To generate the authentication factor of a message that will be broadcast, the vehicle needs to execute one ABE encryption, one point multiplication, one integer multiplication and one hash function.
- 3) To query the vehicle's public key, the managers need to execute a ABE decryption besides the blockchain-related tasks.
- 4) To verify the received message, the vehicle needs to execute one hash function, two point multiplications and one point addition.

To show the practicability of our design, we deploy a software prototype based on the Hyperledger Fabric. To be more specific, we first initialize a Hyperledger network using the Hyperledger-composer V0.20.7 in the AWS on a machine running an Intel Xeon Skylake 6133 (2.5 GHz) with 2 cores and being installed Ubuntu 16.04 OS. The smart contract (written in JavaScript) is deployed over a local area network that is consisted of four nodes.

- The first one is the orderer node that is used to set the environment values, build the configuration tree and local structure of Membership Service Provider, and further start up the core services (e.g., consensus processing).
- The second and third nodes are the peers that are active docker containers (with the version of 18.09.3). Both of them can communicate with other peers on the network and keep a copy of the ledger locally.
- The fourth node acts as a CA which can provide registration of identities and issuance of certificates. It provides both anonymity and unlinkability when an entity executes a transaction on the blockchain network.

All the four nodes deployed have the IP addresses ranging from 192.168.0.1 to 192.168.0.4. After the initialization step of BPAS, we evaluate the time cost of all the phases in BPAS. Table 3 and Figure. 3 show the run-time cost and the total time needed to perform an authentication which is constant. Based on the design of our BPAS, the registration and revocation phases are executed offline by TA, while the login and message signing phase is run by the vehicle which intends to inform others about the traffic condition. The signed messages are authenticated by other vehicles while the public key is queried with the assistance of RSU. The performance demonstrates that the bottleneck of BPAS is in the ABE operations which can be dynamically adjustable by using an "existing and more efficient" scheme. Both the registration and revocation steps yield a few seconds of run-time cost, but they are not performed frequently. Therefore their impact may be limited w.r.t. the operating efficiency.

TABLE 1: Time cost of basic operations (in microsecond)

Operation	Average time
Point addition	0.184
Scalar multiplication	64.99
Integer Multiplication	0.179
Hash	0.296

TABLE 2: Time cost of ABE operations (in microsecond)

Operation	Average time
ABE.TSetup	130.00
ABE.KeyDel	224.19
ABE.ArrtEnc	390.87
ABE.Decrypt	77.02
ABE.TSetup	134.68

8 CONCLUSION

In this paper we have designed a BPAS for vehicular ad-hoc networks, which not only ensures the accuracy and trustworthiness of messages exchanged in VANETs, but also protects vehicle privacy. The novelty of BPAS is the decent integration of blockchain feature and cryptographic primitives, so that it can achieve efficiently privacy-preserving authentication even when trusted party is offline. We've then performed an overall analysis to demonstrate the security requirements that BPAS satisfies. We've also implemented BPAS and evaluated the performance to explore its potential in real-world applications.

As part of our future work, we will extend BPAS to support batch verification of multiple messages to optimize verification and reduce resource consumption. We will also explore flexible privacy preserving mechanisms to address different security needs of VANETs in practice.

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TABLE 3: Running time cost in BPAS (in second)

Step	Running time
Registration (TA)	2.466
Login and message signing (Vehicles)	0.457
Query public key (RSU)	0.302
Message authentication (Vehicles)	0.130
Revocation (TA)	2.338

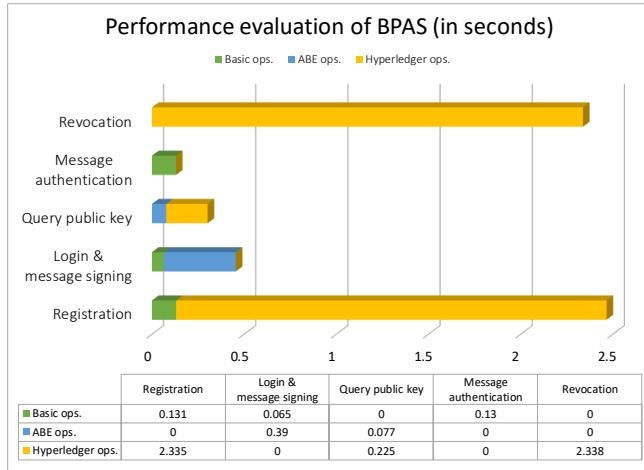


Fig. 3: Performance evaluation of BPAS

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